right atrium and is often used as a surrogate for **right ventricular preload.** Clinically, RAP is equivalent to central venous pressure (CVP) when measured via a central venous catheter. Normal RAP ranges from **2–8 mmHg.**

An elevated RAP suggests impaired right-sided emptying or increased venous return pressure. This can be seen in: Cardiogenic shock (right ventricular failure) Obstructive shock (e.g., pulmonary embolism, cardiac tamponade) Volume overload states. A low RAP suggests decreased venous return, as seen in hypovolemic shock.

CLINICAL REASONING

RAP helps differentiate cardiogenic and obstructive shock from hypovolemic shock.

Pulmonary Capillary Wedge Pressure (PCWP) Pulmonary capillary wedge pressure estimates **left atrial pressure** and therefore reflects **left ventricular preload.** It is obtained using a pulmonary artery catheter, which measures pressure transmitted backward from the left side of the heart through the pulmonary circulation. Normal PCWP ranges from approximately **6–12 mmHg.** Low PCWP suggests reduced preload (e.g., hypovolemia). Elevated PCWP indicates increased left-sided filling pressures, as seen in cardiogenic shock, where blood backs up into the pulmonary circulation due to impaired ventricular pumping.

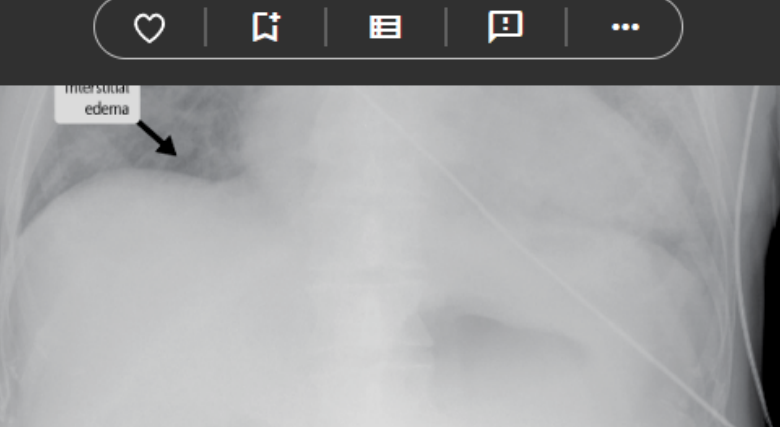


Figure 1 Pulmonary edema

They also usually have pulmonary crackles on chest exam. In patients with pulmonary edema, IV fluids may worsen the hypoxia and potentially cause respiratory failure.

Why is IV fluid an effective first treatment for shock?

How Do We Manage Distributive Shock With

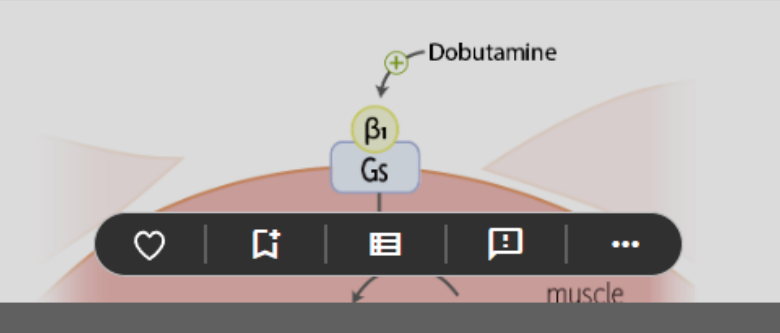
Phenylephrine	+++	None	None	None	None
Epinephrine	+++	++	++	None	None
Vasopressin	None	None	None	+++	None
Dopamine	+	+	+	None	++

Which vasopressor acts only at α_1 adrenergic receptors?

The other class of medications used in treatment of Shock are **inotropes.** The

Inotropes need to be used carefully in the setting of myocardial infarction (MI) or chest pain. This is because both inotropes so strongly stimulate the heart that the energy expended may exceed oxygen supply in patients with coronary ischemia or MI. Additionally, inotropic agents also tend to be chronotropic agents, increasing the risk of tachyarrhythmias that can be treatment-limiting and life-threatening.

- Dobutatmine;Dobutamine strongly stimulates β_1 receptors, causing an **increase in cardiac contractility and heart rate** and, therefore, cardiac output. By weakly stimulating β_2 receptors, dobutamine also causes mild arteriolar vasodilation, reducing SVR and afterload; hence, it is often helpful in treating heart failure. It has minimal effect on α_1 receptors, so causes little or no vasoconstriction. Dobutamine can raise the heart rate substantially and can also precipitate ventricular and atrial arrhythmias through its β_1 stimulation. Because it causes vasodilation, it can sometimes lower blood pressure.
- The inotrope **milrinone** works differently; it inhibits phosphodiesterase 3 ([PDE3] see Figure 2). PDE3 lies downstream, inside smooth muscle cells and on the β_1 pathway; PDE3 degrades cyclic adenosine monophosphate (cAMP) into AMP, which leads to a decrease in intracellular Ca^{2+} and, therefore, a decrease in contractility. By inhibiting PDE3, milrinone decreases cAMP degradation and thus increases myocyte intracellular Ca^{2+} and cardiac contractility. This has the effect of **increasing stroke volume and cardiac output**. By inhibiting PDE3, milrinone also causes vasodilation throughout the vasculature. Milrinone's combined effect of increasing contractility and decreasing afterload makes it great for treating cardiogenic shock.



Mechanical Devices

Some patients with severe cardiogenic shock do not respond to drugs alone and require placement of a **mechanical assist device** to support them until their hearts recover. None of the devices described below have demonstrated clear improvements in patient outcomes, including mortality. These devices should not be used in patients with sepsis, bleeding, or severe aortic disease:

- **Intra-aortic balloon pump (IABP):** This is a catheter-inserted pump that lies in the proximal aorta and coordinates with the ECG to assist the heart in pumping blood forward (Figure 3).
- **Percutaneous left ventricle assist devices:** These require surgical placement and come with a higher risk than IABP. They can be used to support the left ventricle, even when a thrombus is present in-situ, and can also provide support if right ventricular failure occurs. In addition, these devices can be used in the presence of a prosthetic aortic valve or aortic stenosis.
- **Extracorporeal membrane oxygenation (ECMO):** This provides complete cardiopulmonary support for patients with both cardiac and respiratory failure. It involves pumping the patient's blood into a heart-lung machine for ioxide. Full hemodynamic support can be provided by this method. There are currently

Cardiogenic shock: Inotropes are used to treat cardiogenic shock. The two inotropes used in cardiogenic shock are dobutamine and milrinone. As discussed, many adrenergic vasopressors also have strong inotropic activity, most notably epinephrine and dopamine.

CLINICAL CORRELATION

Early assessment of cardiac function is critical in shock evaluation. While formal echocardiography requires a technician, equipment transport, image acquisition, and delayed cardiologist interpretation, Point Of Care UltraSound (POCUS) allows you to assess cardiac function in real-time at the bedside.POCUS reveals the heart's systolic function, measured as ejection fraction (EF). A newly reduced EF in the setting of shock suggests cardiogenic etiology—though this determination requires knowledge of the patient's baseline function. Normal or supranormal EF points toward distributive, hypovolemic, or obstructive shock.

The **pulmonary type of obstructive shock** (from a pulmonary embolus) is treated initially with IV fluids (and pressor drugs if needed), then definitively by dissolving the embolus, often with thrombolytic drugs like tissue plasminogen activators.

The **structural causes of obstructive shock** are treated initially with IV fluids, then with surgical techniques that correct the obstruction:

- Cardiac tamponade is treated by withdrawing pericardial fluid (pericardiocente
- Tension pneumothorax is treated with a thoracostomy tube placed to

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ALICE L WALTON SCHOOL OF MEDICINE

Pericardial Effusion and Cardiac Tamponade

Last updated March 8, 2026 13 min read ScholarRx

Adapted by: Diptiman Bose, Phil Riley, Patrick Brooks, Robert Donnell

Editors: Jordan Finn, Russell Johnson

Adapted from: Pericardial Effusion and Cardiac Tamponade

Listen to this Brick
0:00 / 21:58 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the clinical presentations of pericardial effusion and cardiac tamponade
2. Explain the pathophysiology of pericardial effusion and cardiac tamponade.
3. Interpret the al effusion and

What do you want to learn?

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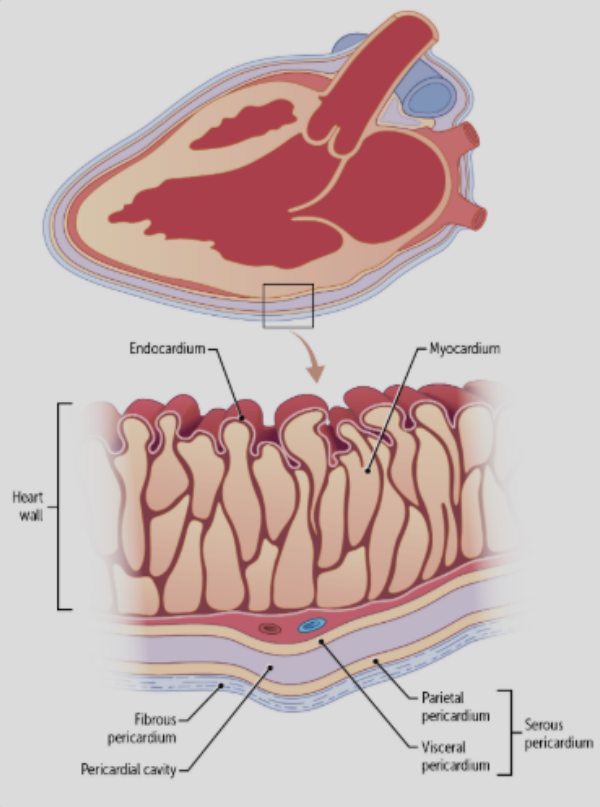
may contain more than 2000 mL of fluid. When a pericardial effusion gets so large that it starts compressing the heart and interfering with its ability to pump blood, we refer to it as **cardiac tamponade**. Cardiac tamponade is a medical emergency, and, if left untreated, can progress to obstructive shock and death.

Pericardial Anatomy

Before delving into more detail, let's briefly review some anatomy so that we can better visualize these disorders. Take a look at Figure 1.

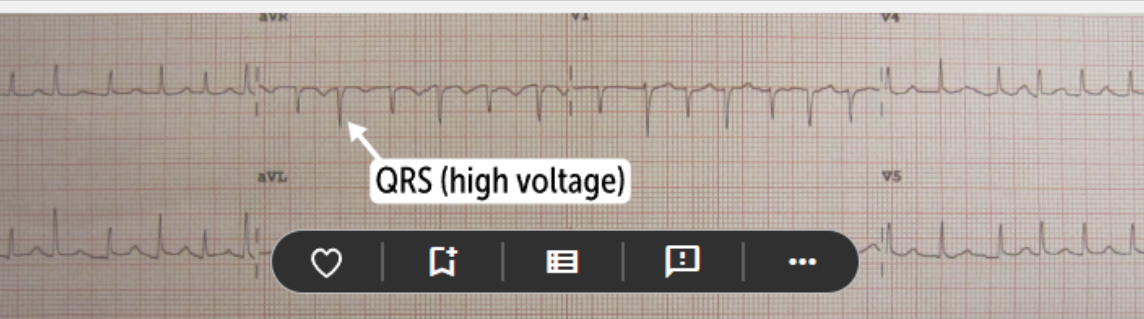


Mnemonic

What are the three D's in Beck's triad of cardiac tamponade?

Electrocardiography

Patients with pericardial effusion (with or without cardiac tamponade) may show a number of findings on their electrocardiogram (ECG). The most specific and striking is **electrical alternans**, a beat-to-beat variation in the QRS complex amplitude (ie, QRS complexes alternate between being low voltage and being high voltage), as shown in Figure 4.



Laboratory Studies

If possible, the underlying cause of the pericardial effusion should be determined. Examples of the many studies commonly done in the workup for pericardial effusion include:

- Complete blood count (looking for leukocytosis as a sign of infection)
- Serum C-reactive protein or erythrocyte sedimentation rate, elevated in inflammatory disorders like lupus
- Serum BUN/creatinine, looking for reduced kidney function
- TSH (elevations show hypothyroidism)

What is the preferred imaging modality to establish the diagnosis of pericardial effusion and the diagnosis of cardiac tamponade?

How Do We Manage Pericardial Effusion and Cardiac Tamponade? (L

Review Questions

1. Which of the following is the most specific ECG finding for cardiac tamponade?

A. ○ Beck triad

B. ● Electrical alternans

C. ○ High-voltage QRS complexes

D. ○ No P waves

E. ○ Pulsus paradoxus

Hide Explanation

The correct answer is electrical alternans (B). This beat-to-beat variation in the QRS complex amplitude (ie, alternating low-voltage and high-voltage QRS complexes) is caused by the "swinging" motion of the heart within the fluid-filled pericardial sac. A large pericardial effusion leads to low-voltage (not high-voltage) QRS complexes (C). Atrial fibrillation (not cardiac tamponade) leads to absent discrete P waves (D). The other responses are signs of effusion that are not ECG findings: pulsus paradoxus (E) is a decrease in systolic blood pressure by >10 mm Hg during inspiration, while the Beck triad (A) consists of hypotension, jugular venous distention, and muffled heart sounds.

B. ○ Myocardial infarction

C. ○ Hypothyroidism

D. ● Osteoporosis

E. ○ Uremia

Hide Explanation

The correct answer is osteoporosis (D). Cardiothoracic surgery (A), myocardial infarction, (B), hypothyroidism (C), and uremia (E) have all been associated with increased risk for pericardial effusion.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the clinical presentations of pericardial effusion and cardiac tamponade~~

02 ☑ ~~Explain the pathophysiology of pericardial effusion and cardiac tamponade.~~

03 ☑ ~~Interpret the clinical findings to diagnose pericardial effusion and cardiac tamponade~~

04 ☑ ~~Outline the treatment plan for pericardial effusion and cardiac tamponade~~

Objective Confidence: 4/4

ALICE L WALTON SCHOOL OF MEDICINE

Development of the Aortic Arches

Last updated February 28, 2026 10 min read ScholarRx

Adapted by: Alexis Gillett

Editors: Russell Johnson, Jordan Finn

Contributor: Nancy Halliday

Reviewers: Diptiman Bose, Helen Ryals, Carrie Elzie

Adapted from: Development of the Aortic Arches

Listen to this Brick
0:00 / 17:17 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. **Analyze** the sequential development and remodeling of the six embryonic aortic arches to trace how each contributes to specific adult arterial derivatives of the head, neck, and thorax.
2. **Apply** embry metric regression of the aortic s of the right and

What do you want to learn?

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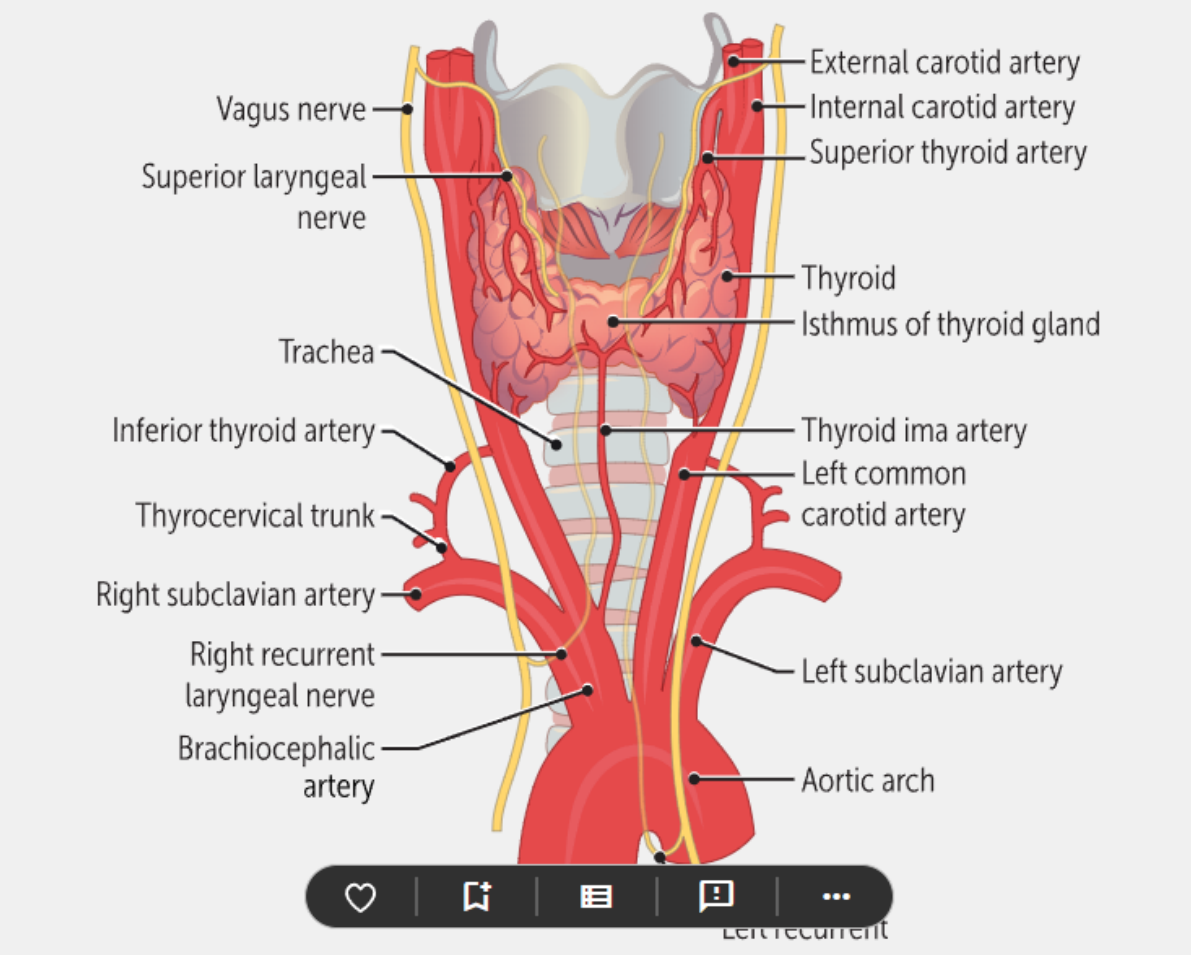
Adult Aortic Anatomy

As shown in Figure 1, the adult aorta gives rise to several major branches whose asymmetric arrangement reflects their embryologic origin. On the **right**, a brachiocephalic trunk divides into the right subclavian and right common carotid arteries. On the **left**, the common carotid and subclavian arteries arise separately from the aortic arch.